Pritish Saha

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Education

Indian Institute of Technology, Kharagpur — **Kharagpur, West Bengal**
B.Tech. in Manufacturing Science & Engineering + M.Tech. in Industrial & Systems Engineering — *Expected completion: May'27*
Relevant Courses: Safety Fundamentals of Generative AI | Operations Research | Probability & Statistics | Linear Algebra

Research & Engineering Experience

Research Fellow (MARS 4.0) | Cambridge AI Safety Hub | UK (Hybrid) | [Report] — ***Dec'25 – Present***
With Prof. Fernando Rosas, studying whether RL improves decisions while retaining memory for belief updates and new goals

- Decoded beliefs from four-layer transformers in fully observed pilots: probe $R^2 = 0.985$–0.997 after 100k training updates
- Built **exact Bayesian filters** and optimal-policy benchmarks for a task where agents pay to observe or stop and guess

Data Science Intern | Axis Bank, Business Intelligence Unit | Mumbai | [Report] — ***May'26 – Jul'26***
Built a PySpark personal-loan fraud pipeline over 224.8M accounts and 1.31B transfer edges, implementing weighted PageRank

- Built a **3.22M-node** scope graph via time-respecting 3-hop BFS, connecting ∼**80%** of September candidate accounts
- Delivered an explainable four-signal review queue: **5.49× mean top-1% lift** across three retrospective monthly cohorts

Research Intern | RAAPID INC | Remote | [Research] [Patent] — ***Mar'25 – Jul'26***
With Prof. Amitava Das, developed parameter-efficient LLM fine-tuning methods and GPU kernels for adapter optimization

- Prototyped **Triton kernels** for covariance estimates, matrix inversion and gradient preconditioning in LoRA fine-tuning
- Integrated **CUDA streams** and events to schedule covariance inversions and Fisher eigensolves, with PyTorch fallbacks

Research Intern | Complex Networks Research Group, IIT Kharagpur | [Report] — ***Jul'25 – May'26***
With Prof. Pawan Goyal, studied latent alignment and reward shaping for mathematical reasoning during LLM post-training

- Added **latent alignment losses** to GRPO for Qwen2.5-Math-7B, using LoRA with 16 rollouts per prompt on one H100
- A selected teacher-augmented run averaged **79.6%** on MATH-500 vs **74.9%** for GRPO (one seed; 8 samples/problem)
- A separate post-hoc audit found shaped credit aligned with GRPO (cosine **0.979**); no shaping signal on all-wrong groups

Publications & Preprints

Mixture of Chapters | Accepted, ICLR 2026 NFAM Workshop | [Code] | [Paper] — ***Mar'26***
Tasmay Pankaj Tibrewal, **Pritish Saha**, Ankit Meda, Kunal Singh, Pradeep Moturi

- Co-developed sparse learnable transformer memory; pretrained **150M–1B-parameter models** on up to **15B tokens**
- Limited ARC-C accuracy loss after instruction tuning to **2.68 pp** versus **6.69 pp** for a compute-matched dense baseline

GRIT: Geometry-aware PEFT | arXiv Preprint | [Code] | [Paper] — ***Jan'26***
Pritish Saha, C. Rajbangshi, R. Goyal, M. Goyal, A. Deo, B. Roy, N. D. Singh, R. Goswami, A. Das

- Scored **66.19% on GSM8K** versus LoRA's 63.38% on Llama-3.1-8B with **27.8% fewer** effective adapter parameters
- Designed a LoRA method with **rank-space preconditioning**, Fisher-guided adapter reprojection and adaptive ranks

Competitions

3rd Place | General Championship Data Analytics, IIT Kharagpur | [Report] — ***Mar'26***
Captained Patel Hall's team to build a conversational analytics dashboard for database queries and automated KPI discovery

- Built text-to-SQL with schema retrieval and repair: **90% execution accuracy** on a 50-query BIRD mini-dev sample
- Automated KPI discovery in a nine-stage LangGraph pipeline; generated **874K synthetic rows** across 58 client profiles

4th Place | ISRO GeoNLI, Inter IIT Tech Meet 14.0 | [Report] — ***Oct'25 – Dec'25***
Co-designed a remote-sensing assistant for captioning, grounding and question answering across RGB, SAR and IR imagery

- Designed a Qwen3-VL-8B curriculum with **LoRA SFT** followed by **DPO** for satellite captioning and question answering
- Paired SAM3 with pyramidal tiling and VLM refinement to count objects, estimate areas and answer spatial queries

National Finalist | American Express Campus Challenge, Decision Science | [Report] — ***Jul'25***
Built a personalized offer-ranking system from customer histories, combining gradient-boosted trees with a residual transformer

- Scored **0.59 MAP@7** on final evaluation; trained a transformer with listwise loss to correct boosted-tree ranking errors
- Generated **3K+ customer-history features** with point-in-time joins and customer-grouped validation to limit leakage

Technical Skills

Languages & Systems: Python, C++, SQL, CUDA/PTX, Triton, PySpark, Hadoop/HDFS, Linux, Docker, SLURM, Git
ML & Post-training: PyTorch, JAX, Transformers, LoRA/QLoRA, DPO, TRL, vLLM, FSDP/DTensor, FlashAttention-2

Extra-Curricular Activities

Karate black belt and tournament winner | Represented Patel Hall in interhall football and water polo | NSS village volunteer